

Lab 02 Results Summary — Bernales

FITTED PARAMETERS

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c (asymptote)      : 1.4938 ± 0.0018
                   t = 811.46, p = 0.0000
a (amplitude)      : 3.3290 ± 0.0033
                   t = 1020.13, p = 0.0000
k (growth rate)    : 0.3844 ± 0.0009
                   t = 409.78, p = 0.0000
t0 (inflection)   : 12.4318 ± 0.0064
                   t = 1944.76, p = 0.0000
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GOODNESS OF FIT

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SSE (Sum of Squared Errors): 0.003958
SST (Total Sum of Squares): 152.425688
R2 (Coefficient of Determination): 0.999974
n (number of samples): 96
p (number of parameters): 4
df (degrees of freedom): 92
s (Std. Error of Estimate): 0.006559 m
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AREA UNDER THE CURVE

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Integration limits: 0.0 to 23.8 hours
Area (quad): 73.194111 m·h
Error estimate: 0.000000
Area (trapezoid, fitted): 73.194140 m·h
Area (trapezoid, raw): 73.194733 m·h
Units: m·h (meters × hours)
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READING OF THE RESIDUALS

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The residuals are randomly scattered around zero with no obvious trend or pattern, suggesting the logistic model adequately captures the underlying behavior. The largest residuals occur during the steepest portion of the curve, which is expected as the rounding error ( $\pm 1$  cm) becomes more noticeable when the slope is steep. The residual magnitude (typically  $< 0.02$  m) is consistent with the logger's rounding precision, indicating the fit is appropriate for the given data.
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